

Recent Research Projects

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WildChat

2023–2024

Real user–chatbot interactions were not public

PUBLIC DATASETS

curated prompts

Clean tasks written by annotators or generated by models.

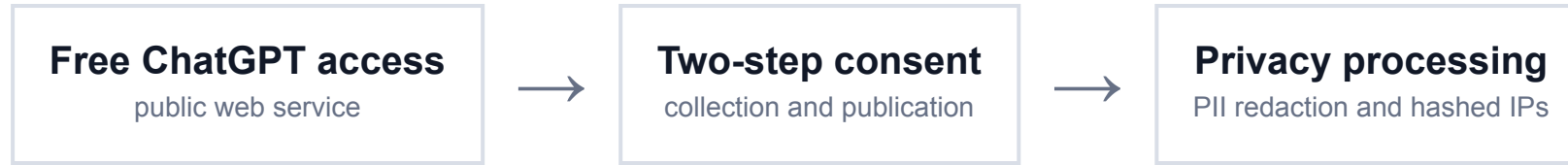
DEPLOYED SYSTEMS

real conversations

Messy, multi-turn, multilingual, and mostly proprietary.

We wanted a public dataset that showed how people actually use ChatGPT.

Users explicitly opted in to data collection



The service did not require an account or personal information.

SCALE

1.0M

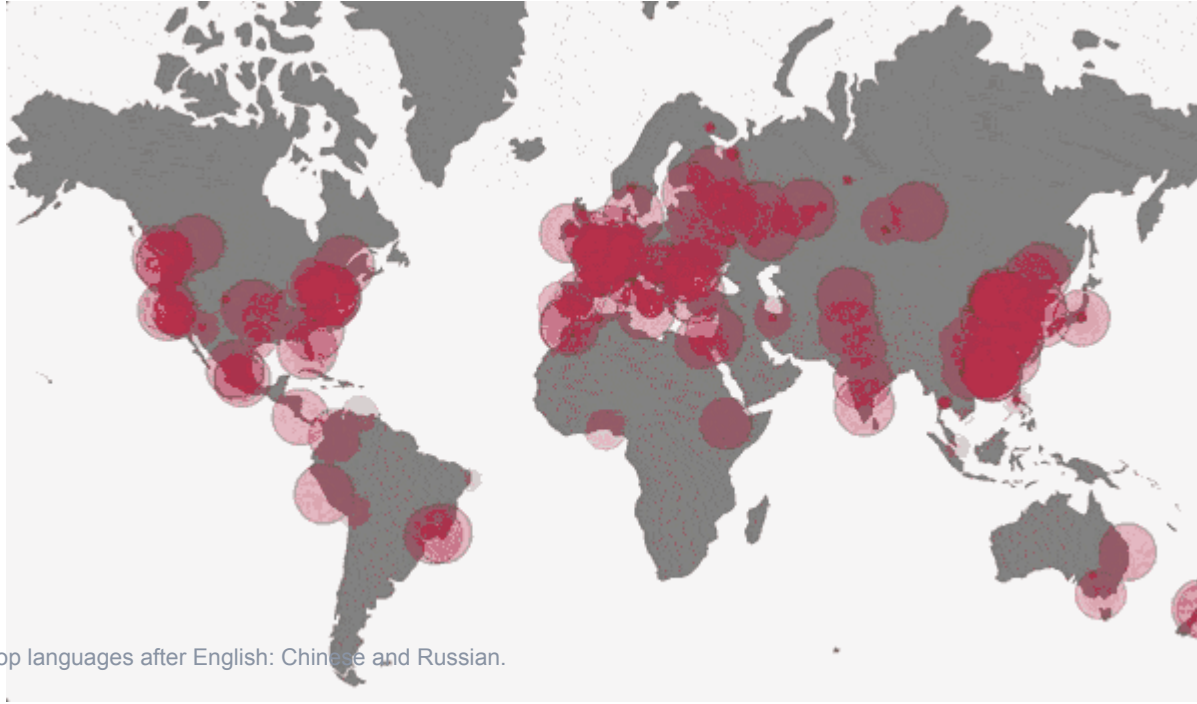
conversations in the ICLR 2024 release

4.8M

conversations in the current public release

Zhao et.al., "WildChat: 1M ChatGPT Interaction Logs in the Wild," ICLR 2024 Spotlight.

Only 53% of turns are in English



68
languages

Real prompts are often underspecified

“Buying a car from a junkyard that hasn't run since 1975...”

“论文的 introduction 怎么写?”

“Why is this invoice absent even though revenue was posted correctly?”

Ambiguity, code-switching, and domain-specific context are common.

Different toxicity detectors find different cases



Turn-level analysis in the WildChat paper. Percentages are detector flags, not ground-truth labels.

The two detectors agree on only 3.73% of user turns.

DOWNSTREAM USE

WildChat became evaluation data

OpenAI o1

safety evaluation

Claude 3

refusal evaluation

WildBench

real-user benchmark

The dataset also supports research on privacy, misuse, multilinguality, and user behavior.

Million-scale chat logs need an interactive viewer

Filter by language, country, toxicity, or text

Search individual conversations

Explore conversations in embedding space

Deng et al., "WildVis," EMNLP 2024 Demo · wildvisualizer.com



Implicit Chain of Thought

2023–2024

Explicit chain of thought serializes every step

$12 \times 3 = ?$

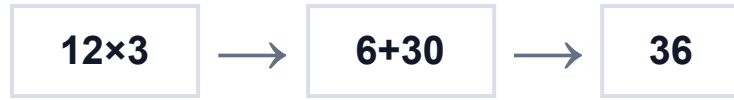
$6 + 30$

36

Longer reasoning means more generated tokens and more inference time.

Reasoning does not have to be expressed in human language

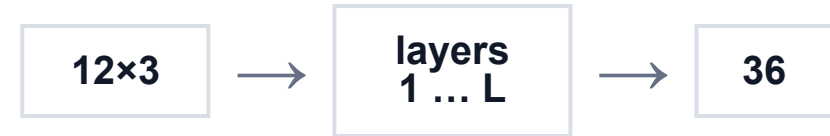
EXPLICIT COT



reason across generated tokens

Deng et al., "Implicit Chain of Thought Reasoning via Knowledge Distillation," 2023.

IMPLICIT COT



reason through hidden states

Implicit CoT solves 5×5 multiplication directly



Exact match; task-specific training on augmented multiplication data; batch size 1.
Implicit CoT: 4.3 examples/s · Explicit CoT: 0.8 examples/s

Remove a small number of reasoning tokens

Finetune and repeat

Question: 1 2 3 4 5 6 7 8 9 × 9 8 7 6 5 4 3 2 1 = ?

Remove 0 CoT Token(s) & Finetune

Speed



× 1.0

Accuracy



100%

Chain of Thought (CoT):

```

      1 2 3 4 5 6 7 8 9
    × 9 8 7 6 5 4 3 2 1
    -----
      1 2 3 4 5 6 7 8 9
      2 4 6 9 1 3 5 7 8
      3 7 0 3 7 0 3 6 7
      4 9 3 8 2 7 1 5 6
      6 1 7 2 8 3 9 4 5
      7 4 0 7 4 0 7 3 4
      8 6 4 1 9 7 5 2 3
      9 8 7 6 5 4 3 1 2
      1 1 1 1 1 1 1 0 1
    -----
    1 2 1 9 3 2 6 3 1 1 1 2 6 3 5 2 6 9

```

Answer: 1 2 1 9 3 2 6 3 1 1 1 2 6 3 5 2 6 9

LIMITATION

Hidden reasoning is harder to inspect

BENEFIT

fewer output tokens

COST

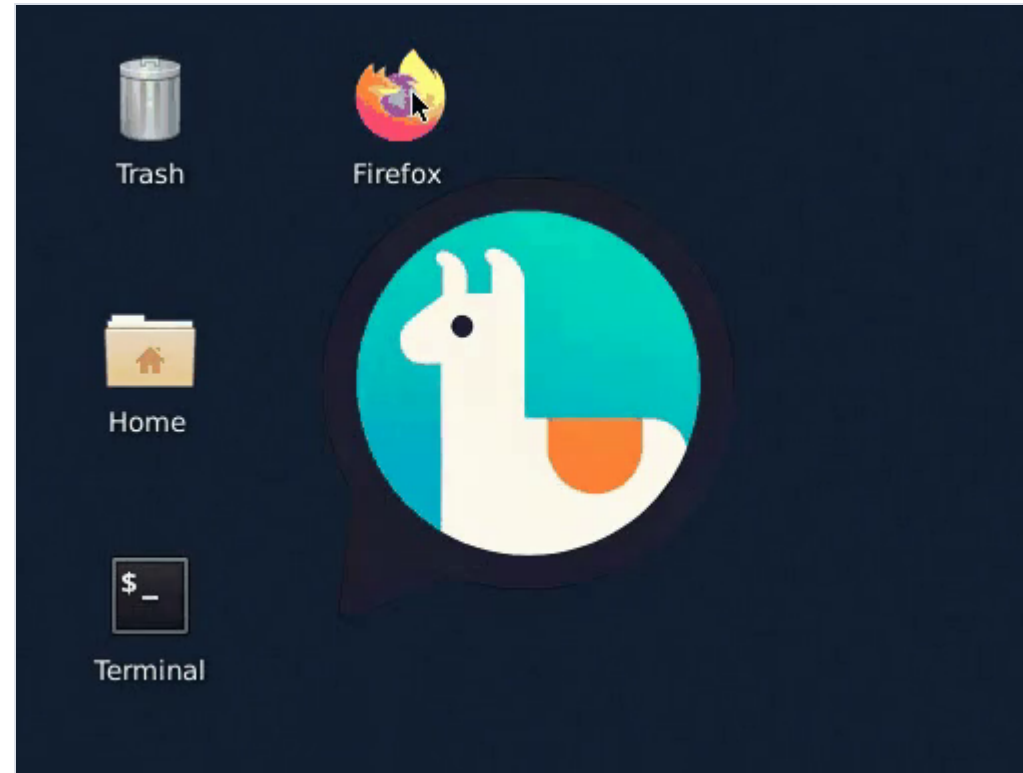
less interpretability

Implicit CoT is a speed–transparency trade-off, not a universal replacement for explicit reasoning.

NeuralOS

2024–2026

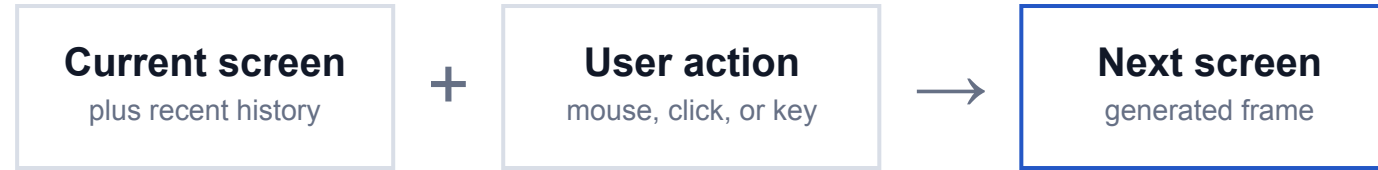
Every screen pixel is generated by a neural network



Rivard et al., "NeuralOS," ICLR 2026.

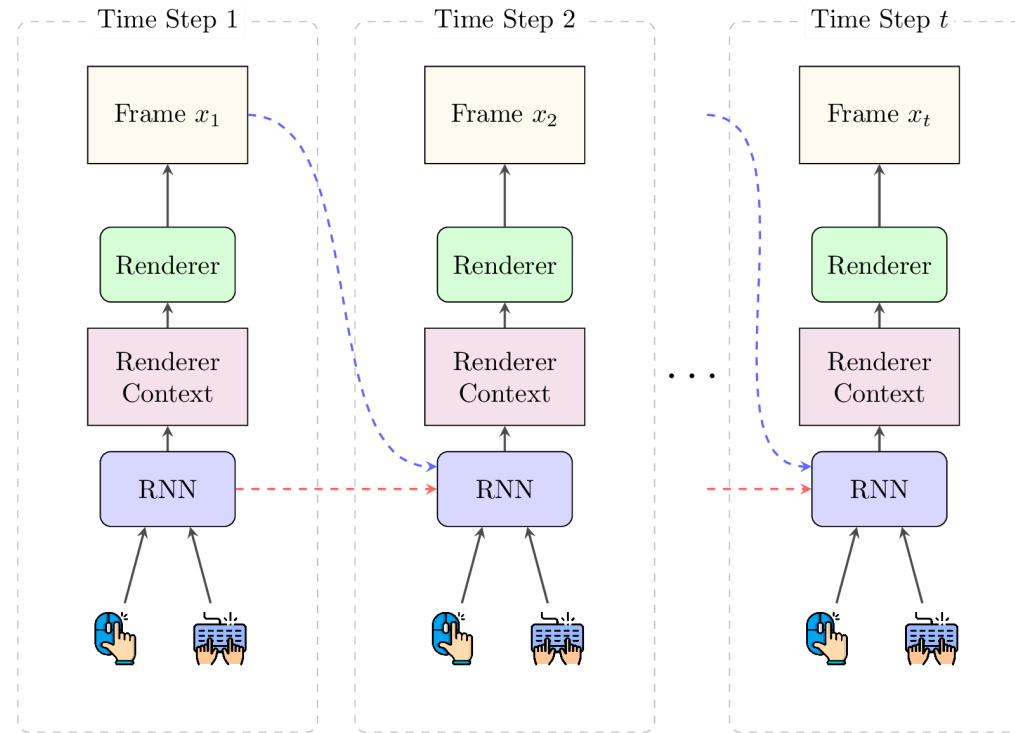
No real kernel · no installed applications · no fixed UI tree

Predict the next frame after each user action



Feed the generated frame back into the model and repeat.

An RNN stores state; diffusion renders the screen



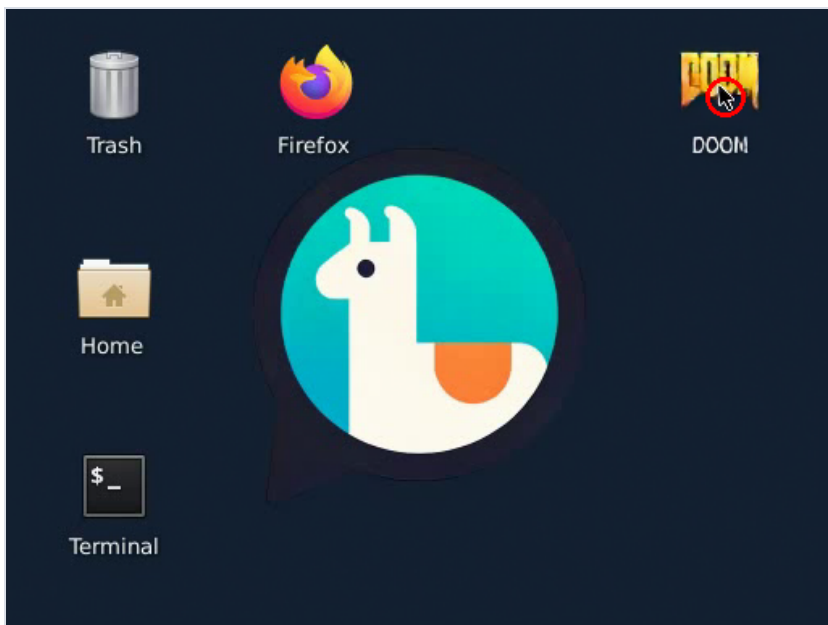
RNN
tracks long-term OS state

DIFFUSION MODEL
renders the next frame

Random exploration prevents action shortcuts



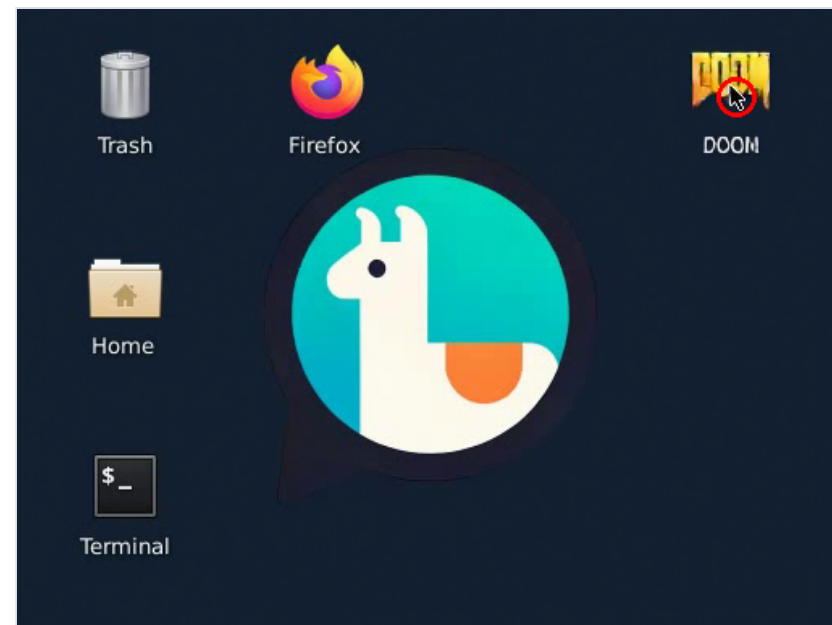
Doom was learned without being installed



generated desktop



double-click launches Doom



escape returns to desktop

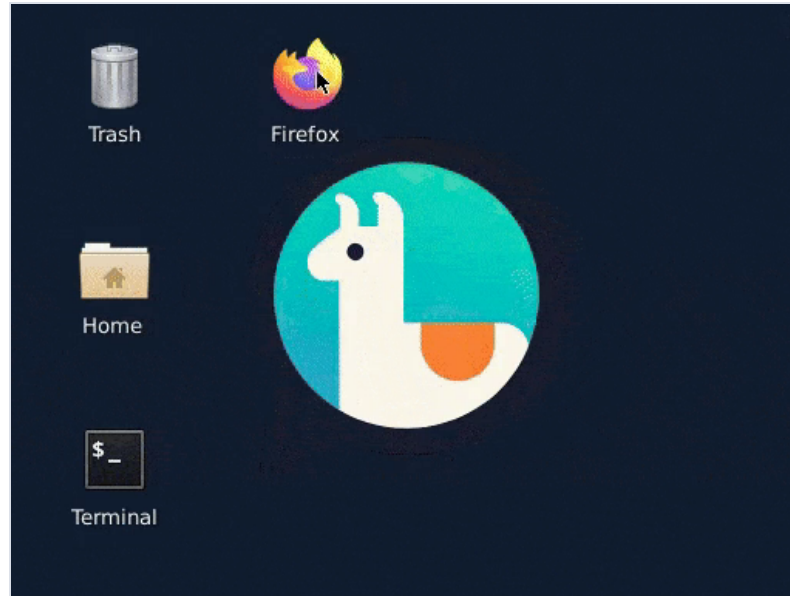
Training videos were fabricated by adding an icon and splicing in gameplay.

Resolution increased fourfold in 18 months

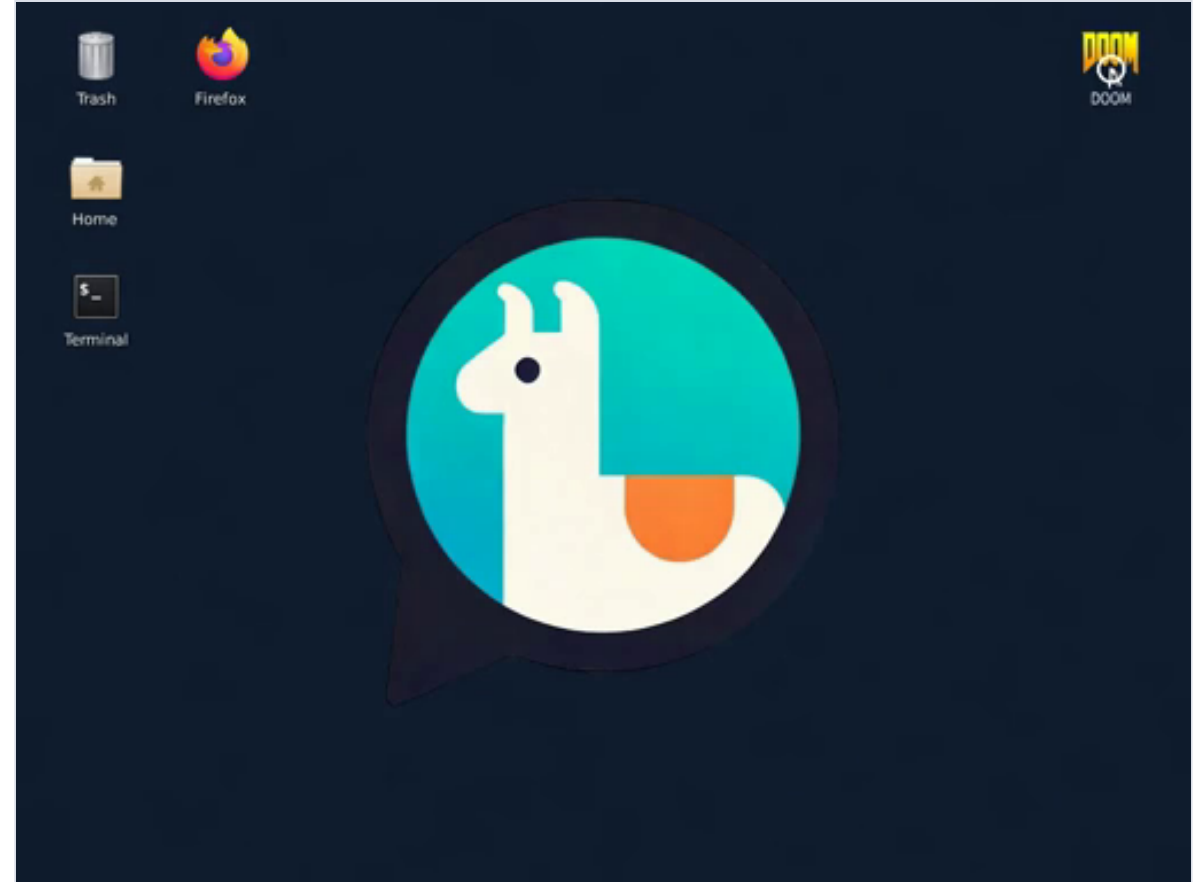
2024 · 256×256



2025 · 512×384



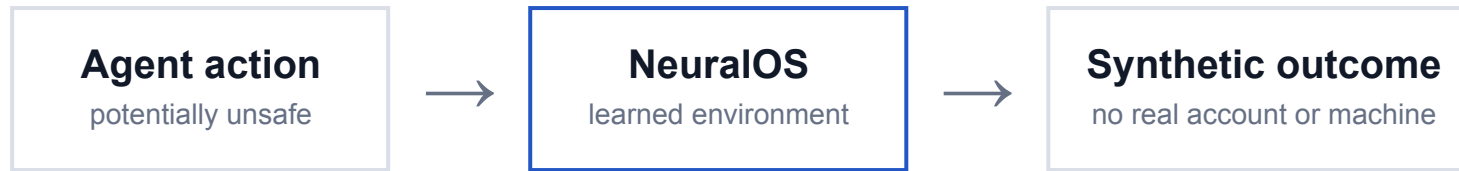
2026 · 1024×768



neural-os.com

USE CASE

NeuralOS can be a simulator for computer-use agents



A computer-use analogue of a driving simulator.

Program-as-Weights

2025–2026

MOTIVATION

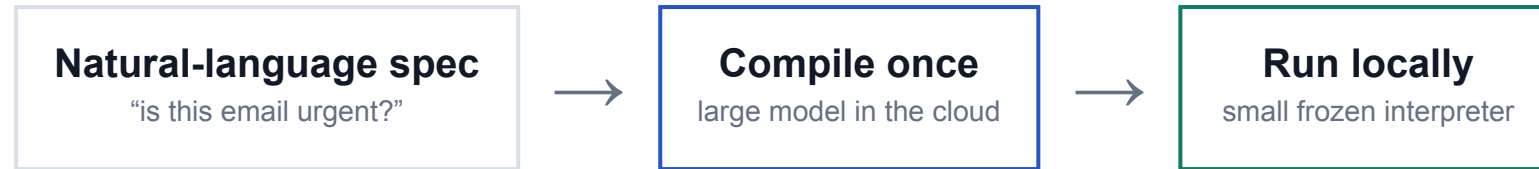
Some useful functions are easy to describe but hard to code

“Is this email urgent?”

Rules and regular expressions are brittle; a full LLM call is often excessive.

APPROACH

Compile once; run locally



After compilation: no API key, no internet, and no repeated large-model call.

One interpreter can run many programs



COMPILER

4B

used once per program

LOCAL RUNTIME

0.6B + 23MB

shared interpreter plus one adapter

The result is called like a normal function

```
# Compile and load a local function.  
fn = paw.compile_and_load(  
    "Classify if an email requires immediate attention"  
)  
  
# This call runs locally.  
fn("Need your signature by EOD!") # → "immediate"
```

PAW on 0.6B reaches 73.8% on FuzzyBench



FuzzyBench verified test set; exact match.

About 50× less model memory than the 32B baseline

COMPOSITION

One product can use many small programs

30

compiled programs in the site helper

28

serve live in routing, retrieval, answering,
and validation

programasweights.com/#ask

A DIRECTION I AM INTERESTED IN

AI systems that can be customized to particular users and use cases

General models are useful; many applications also need behavior tailored to their data, constraints, and deployment setting.

Thank you